KNECK OFFICIAL

Mathematics — Paper 1

121/1 • ORIGINAL REVISION SET 01 • 2 hours 30 minutes • 100 marks

Name: _____ Class: _____ Date: _____

Independent KCSE-style practice. Not issued, endorsed or predicted by KNEC. These are original questions, not leaked or past official papers. Component numbering is checked against the KNEC 2026 timetable. A teacher should review these practice materials before formal classroom assessment.

CANDIDATE INSTRUCTIONS

Answer ALL questions in Section A and ANY FIVE in Section B. Section A: 50 marks; Section B: 50 marks. Use a non-programmable calculator and geometrical instruments. Take π = 22/7 where needed.

Use a separate answer booklet. Show working where appropriate. The marking guidance begins after the END OF QUESTIONS page; do not consult it until you finish. Practical and project components require the stated equipment and teacher supervision, not merely a printed worksheet.

SECTION A

Answer all 16 questions.

1. Evaluate (3/4 - 2/5) ÷ 7/10. [3 marks]

2. Simplify $(6x^2y \times 3xy^3)/(9x^2y^2)$, where x,y are non-zero. [3 marks]

3. Solve 3(2x - 1) - 4 = 2x + 9. [3 marks]

4. Write 0.0000728 in standard form; then round it to two significant figures. [3 marks]

5. A trader buys an item for KSh 2,400 and sells it at a profit of 15%. Find the selling price. [3 marks]

6. Find the gradient and equation of the line through (2,3) and (6,11). [3 marks]

7. Solve simultaneously 2x + y = 11 and x - y = 1. [3 marks]

8. Find the interior angle and the number of diagonals of a regular octagon. [3 marks]

9. Convert 72 km/h to m/s; find the distance travelled at this speed in 45 seconds. [3 marks]

10. Factorise $2x^2 - 5x - 3$ completely. [3 marks]

11. A bag has 5 red, 3 blue and 2 green balls. Find P(not blue) and P(red or green). [3 marks]

12. The scores 4, 6, 7, 7, 11 have frequencies 1 each. Find the mean, median and mode. [3 marks]

13. The bearing of B from A is 065°. Find the bearing of A from B; state the acute angle AB makes with the east direction. [3 marks]

14. A cylinder has radius 3.5 cm and height 10 cm. Find its volume. [3 marks]

15. Solve and represent on a number line: $2x - 3 < 7$ and $x + 4 \geq 2$. [4 marks]

16. Using ruler and compasses only, construct triangle ABC with AB = 6 cm, AC = 5 cm and BC = 4 cm. Construct the perpendicular bisector of AB. [4 marks]

SECTION B

Answer any FIVE of the following eight questions. Each question carries 10 marks.

17. A rectangle has length (x + 5) cm and width x cm. Its area is 84 cm². (a) Form a quadratic equation. (2) (b) Solve it and state the dimensions. (4) (c) Find the perimeter and length of a diagonal, correct to 2 decimal places. (4) [10 marks]

18. A map scale is 1:50,000. A straight road measures 8.4 cm on the map. A rectangular farm measures 3.2 cm by 2.5 cm. (a) Find the actual road length in km. (3) (b) Find the farm area in hectares. (4) (c) A cyclist takes 14 minutes along the road. Find average speed in km/h. (3) [10 marks]

19. The table of scores and frequencies is: scores 2,4,6,8,10; frequencies 3,5,6,4,2. (a) Calculate the mean. (4) (b) Find the median. (2) (c) Draw a bar chart with labelled axes. (4) [10 marks]

20. A bus leaves Nairobi at 08:20 and travels 180 km at 60 km/h, stops for 25 minutes, then travels 120 km at 80 km/h. (a) Calculate arrival time. (4) (b) Calculate overall average speed including the stop. (3) (c) Find fuel cost at 8 km per litre and KSh 180 per litre. (3) [10 marks]

21. A ladder 10 m long leans against a vertical wall. Its foot is 6 m from the wall. (a) Find height reached. (3) (b) Find the angle with the ground. (3) (c) If the foot moves to 8 m from the wall, find the decrease in height. (4) [10 marks]

22. For vectors a = (3,2) and b = (-1,4), calculate (a) 2a - b (3), (b) |a + b| (3). Points A(1,2), B(7,6): find midpoint M and vector AM. (4) [10 marks]

23. A tank is filled by tap A in 6 hours and by tap B in 8 hours. A leak empties a full tank in 24 hours. (a) Find the fraction filled in one hour with both taps and the leak open. (4) (b) Find time to fill an empty tank. (3) (c) Find time using A alone with the leak open. (3) [10 marks]

24. A shop mixes 12 kg of tea costing KSh 300/kg with 8 kg costing KSh 450/kg. (a) Find total cost and mean cost/kg. (4) (b) Find selling price/kg for 20% profit on cost. (3) (c) Find percentage profit if sold at KSh 450/kg. (3) [10 marks]

END OF QUESTIONS

MARKING GUIDANCE

Mathematics • Paper 1 • Set 01

Suggested answers and assessment guidance, not an official marking scheme. Accept equivalent correct reasoning, valid examples and properly justified alternative methods. Marks for an item must not exceed its printed allocation. For essays use relevant textual evidence and the rubric; no plot summary alone.

SECTION A

1. (3 marks) 3/4 - 2/5 = 7/20; multiplying by 10/7 gives 1/2. Award one mark for subtraction, reciprocal and final answer.

2. (3 marks) $18x^3y^4/(9x^2y^2) = 2xy^2$. Award coefficient, x exponent and y exponent.

3. (3 marks) 6x - 7 = 2x + 9; 4x = 16; x = 4.

4. (3 marks) 7.28 × 10^-5; 7.3 × 10^-5 (or 0.000073). 1 for standard form, 2 for rounding.

5. (3 marks) Profit = 360; selling price = KSh 2,760. 1 for percentage conversion, 1 for profit, 1 final.

6. (3 marks) m = (11-3)/(6-2) = 2; y - 3 = 2(x - 2); y = 2x - 1.

7. (3 marks) Adding gives 3x = 12; x = 4 and y = 3.

8. (3 marks) Interior angle = 135°; diagonals = n(n-3)/2 = 20. Award 1 for angle method, 1 angle, 1 diagonals.

9. (3 marks) 72 × 1000/3600 = 20 m/s; distance = 900 m. Conversion method 1, speed 1, distance 1.

10. (3 marks) (2x + 1)(x - 3). Award splitting middle term, grouping and factors.

11. (3 marks) Both probabilities are 7/10. 1 for total 10, 1 each answer.

12. (3 marks) Mean = 35/5 = 7; median = 7; mode = 7. One mark each.

13. (3 marks) 245°; 25°. Award 1 for reciprocal-bearing method, 1 each result.

14. (3 marks) $V = \pi r^2h = (22/7)(3.5^2)(10) = 385$ cm³. Formula, substitution, answer.

15. (4 marks) $x < 5$; $x \geq -2$; intersection $-2 \leq x < 5$. Closed dot -2, open dot 5 with shaded interval. One mark per stage.

16. (4 marks) AB to scale (1), intersecting arcs at correct radii and C (1), joining AC and BC (1), correct perpendicular bisector with construction arcs (1).

SECTION B

17. (10 marks) $x(x+5)=84$, $x^2+5x-84=0$ (2). $(x+12)(x-7)=0$; reject -12; width 7, length 12 (4). Perimeter 38 cm (2); diagonal $\sqrt{193} = 13.89$ cm (2).

18. (10 marks) Road 4.2 km (3). Actual sides 1.6 km and 1.25 km; area 2 km² = 200 ha (4). Speed = 4.2/(14/60) = 18 km/h (3).

19. (10 marks) $\Sigma f=20$, $\Sigma fx=114$; mean 5.7 (4). 10th and 11th observations are 6 so median 6 (2). Chart: sensible scale, labelled axes, separated bars and correct heights (4).

20. (10 marks) First leg 3 h; second 1.5 h; elapsed 4 h 55 min; arrival 13:15 (4). 300/(59/12)=3600/59 = 61.02 km/h (3). 37.5 L; KSh 6,750 (3).

21. (10 marks) Height √(100-36)=8 m (3). Angle cos^-1(6/10)=53.13° (3). New height √(100-64)=6 m; decrease 2 m (4).

22. (10 marks) 2a-b=(7,0) (3). a+b=(2,6); magnitude √40=2√10 (3). M=(4,4); AM=(3,2) (4).

23. (10 marks) 1/6+1/8-1/24=1/4 (4). Four hours (3). 1/6-1/24=1/8, so eight hours (3).

24. (10 marks) Cost 3,600+3,600=7,200; mass 20 kg; mean 360/kg (4). Selling price 432/kg (3). Profit 90/kg; 90/360 ×100=25% (3).

REVISION CHECK

Mark your work, list three misconceptions, revise those topics and attempt the incorrect questions again without the guide. These topics are revision priorities, not a forecast of the real examination.

Paper/component reference: https://www.knec.ac.ke/wp-content/uploads/2026/02/2026-KCSE-Timetable.pdf